Exposing model calls as REST endpoints — setup guide

Goal

Make the demo's models available to external SaaS providers as plain REST endpoints, in **both** API formats, through the Privilege LLM gateway.

Format	Endpoint	Auth	Status (2026-11-13)
GenAI (Anthropic Messages)	`POST https://mcpgw.ai-demo.ping-devops.com/llm/anthropic/v1/messages`	virtual key as `x-api-key` + `anthropic-version: 2023-06-01`	■ verified 200
OpenAI (chat completions, keyless lane)	`POST https://ai-demo.ping-devops.com/llm-gw/v1/chat/completions`	none (public ingress)	■ verified 200, model `gpt-oss-20b`
OpenAI (governed, gateway virtual key)	`POST https://mcpgw.ai-demo.ping-devops.com/llm/openai/v1/chat/completions`	virtual key as `Authorization: Bearer`	route correct; needs a live OpenAI-bound key
GenAI (Google native)	`POST https://mcpgw.ai-demo.ping-devops.com/llm/google/v1beta/models/gemini-2.5-flash:generateContent`	virtual key as `Authorization: Bearer`	key valid; upstream Google key invalid/billing depleted

The gateway does **not** translate formats: use the route that matches the key's provider. Each virtual key is bound to exactly one provider — wrong route returns 403 `wrong_provider`.

Setup: create a virtual key (console)

1. Sign in to the Privilege console (`https://local.procyon.ai:8643`).
2. **Agent Registry** → **virtual keys** → create a key:
- Pick the **provider** (Anthropic / OpenAI / llama.cpp→saved as "custom" / Google). The provider's real API key must be in hand — Privilege stores it and injects it server-side; the caller's key is never forwarded.
- Add the **allowed models** list (exact strings; e.g. `claude-haiku-4-5-20251001`, `gpt-4o-mini`, `meta.llama3-2-3b-instruct-v1:0`). A model not on the list is 403.
- Set request/token rate limits, budget, expiry as needed.
3. Save. New keys are picked up **immediately**; edits to a key take ~4 minutes.
4. View the saved key and copy the `sk-orion-...` value — it is shown once.

Known upstream gotcha: the gateway forwards to a custom (llama.cpp) upstream from inside its own pod, so the CustomUpstream BaseURL must be a URL the gateway pod can reach (`https://ai-demo.ping-devops.com/llm-gw`), not a loopback.

Usage

GenAI format (1 curl, 1 script)

```
# curl — Anthropic Messages format
curl https://mcpgw.ai-demo.ping-devops.com/llm/anthropic/v1/messages \
  -H "x-api-key: <virtual-key>" \
  -H "anthropic-version: 2023-06-01" \
  -H "content-type: application/json" \
  -d '{"model":"claude-haiku-4-5-20251001","max_tokens":8,"messages":[{"role":"user","content":"Reply with the singl

# script (reads key from VIRTUAL_KEY.txt)
./privilege/scripts/test-genai.sh        # VERIFIED 200
```

SDK path for SaaS tenants: point the Anthropic SDK's base URL at

`https://mcpgw.ai-demo.ping-devops.com/llm/anthropic` and pass the virtual

key as the API key.

OpenAI format (1 curl, 1 script)

```
# curl — OpenAI chat completions format, keyless lane (our llama.cpp endpoint)
curl https://ai-demo.ping-devops.com/llm-gw/v1/chat/completions \
  -H "content-type: application/json" \
  -d '{"model":"gpt-oss-20b","max_tokens":32,"messages":[{"role":"user","content":"Reply with the single word ok"}]}

# script — no setup needed
./privilege/scripts/test-openai.sh       # VERIFIED 200
```

Governed OpenAI-format variant (virtual key, gateway route):

```
curl https://mcpgw.ai-demo.ping-devops.com/llm/openai/v1/chat/completions \
  -H "Authorization: Bearer <virtual-key>" \
  -H "content-type: application/json" \
  -d '{"model":"gpt-4o-mini","max_tokens":8,"messages":[{"role":"user","content":"ok"}]}'
```

SDK path for SaaS tenants: point the OpenAI SDK's base URL at

`https://mcpgw.ai-demo.ping-devops.com/llm/openai/v1` (or the keyless

`https://ai-demo.ping-devops.com/llm-gw/v1`).

Streaming

Both lanes stream: add `"stream":true` (OpenAI format) or `"stream":true`

(Anthropic format returns SSE `message_start`/`content_block_delta` events).

Verified live on the Anthropic route 2026-11-13.

Key file

`VIRTUAL_KEY.txt` (repo root) holds the virtual keys, one per line, with

comments recording each key's provider, route, and verification status.

`./privilege/scripts/test-llm-rest.sh` runs the full matrix against all lanes.

Failure reference (live-verified 2026-11-13)

Symptom	Cause	Fix
401 `virtual key expired or revoked`	key inactive in console	reinstate/recreate key
403 `wrong_provider`	key bound to another provider	use the key's own route
403 `model "..." not allowed for this key`	model not on key's list, or typo	add exact model string; wait ~4 min for edits
404 page not found on `/llm/google/v1/...`	Google lane is `/v1beta` native, not `/v1`	use `.../v1beta/models/<model>:generateContent`
404 page not found on `/llm/custom/...`	CustomUpstream unreachable from gateway pod	set BaseURL to `https://ai-demo.ping-devops.com/llm-gw`
400 `API key not valid` from googleapis.com	RealKey under the Google key is dead	re-enter a valid Google AI Studio key in the console
402 `prepayment credits are depleted`	AI Studio billing empty	top up at ai.studio/projects